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Pirtobrutinib Versus Ibrutinib in Treatment-Naïve and Relapsed/Refractory Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma

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ABSTRACT

PURPOSE Pirtobrutinib, a highly selective, noncovalent Bruton tyrosine kinase inhibitor (BTKi), has shown efficacy and safety in patients with chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL) who received prior covalent BTKi. We report results, to our knowledge, from the first randomized head-to-head comparison of pirtobrutinib versus ibrutinib in BTKi-naïve CLL/SLL in both treatment-naïve (TN) patients and patients with relapsed/refractory (R/R) disease.

PATIENTS AND METHODS Patients (N = 662) were randomly assigned 1:1 to receive pirtobrutinib or ibrutinib. All patients were BTKi-naïve. Primary end points were overall response rate (ORR) by independent review committee (IRC) among all randomly assigned patients (intention to treat [ITT]) and in patients with R/R disease.

RESULTS The study met its primary end points, demonstrating statistically significant noninferiority (NI) of IRC-ORR for pirtobrutinib versus ibrutinib in both the ITT (87.0% [95% CI, 82.9 to 90.4] *v* 78.5% [95% CI, 73.7 to 82.9]; ORR ratio = 1.11 [95% CI, 1.03 to 1.19]; two-sided $P < .0001$) and R/R populations (n = 437; 84.0% [95% CI, 78.5 to 88.6] *v* 74.8% [95% CI, 68.5 to 80.4]; ORR ratio = 1.12 [95% CI, 1.02 to 1.24]; two-sided $P < .0001$). In TN patients (n = 225), IRC-ORR was 92.9% (95% CI, 86.4 to 96.9) with pirtobrutinib versus 85.8% (95% CI, 78.0 to 91.7) with ibrutinib. Investigator assessed ORR results were consistent. Investigator-assessed progression-free survival (PFS) favored pirtobrutinib in the ITT (hazard ratio [HR], 0.57 [95% CI, 0.39 to 0.83]), R/R (HR, 0.73 [95% CI, 0.47 to 1.13]), and TN (HR, 0.24 [95% CI, 0.10 to 0.59]) populations. Cardiac adverse event rates of atrial fibrillation/flutter and hypertension were lower with pirtobrutinib.

CONCLUSION Pirtobrutinib demonstrated NI of ORR versus ibrutinib, with a favorable early PFS trend, particularly in TN patients, and a favorable safety profile including low rates of atrial fibrillation and hypertension.

ACCOMPANYING CONTENT

Listen to the podcast by Dr Friedberg, Dr Woyach, Dr Jurczak, and Dr Davids at ascopubs.org/do/jco-ash-2025-pirtobrutinib-untreated-cll

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Data Supplement

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INTRODUCTION

The covalent Bruton tyrosine kinase inhibitors (cBTKi) ibrutinib, acalabrutinib, and zanubrutinib have improved outcomes and transformed the therapeutic landscape for patients with chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL). Despite these advances, cBTKi share pharmacologic limitations, including low oral bioavailability, short half-life, and tolerability challenges, that may collectively affect sustained BTK inhibition, adherence, and consequently efficacy.[1-3] Ibrutinib was the first approved cBTKi having demonstrated benefit in both treatment-naïve (TN) and relapsed/refractory (R/R) CLL. Subsequently, second-generation cBTKi, acalabrutinib and zanubrutinib, were developed to improve the safety profile of ibrutinib by enhancing selectivity for BTK. Both were investigated in head-to-head studies against ibrutinib in patients with R/R CLL/SLL. Acalabrutinib demonstrated

CONTEXT

Key Objective

Here, we report, to our knowledge, the first randomized phase III clinical trial evaluating the efficacy and safety of a noncovalent Bruton tyrosine kinase inhibitors (BTKi), pirtobrutinib, with a covalent BTKi, ibrutinib, in treatment-naïve patients and patients with BTKi-naïve relapsed/refractory (R/R) CLL/SLL. The primary objective of BRUIN CLL-314 was to establish noninferiority (NI) of independent review committee (IRC)–assessed overall response rate (ORR) with pirtobrutinib compared with ibrutinib in the intention-to-treat (ITT) and R/R populations.

Knowledge Generated

In patients with CLL/SLL who are BTKi-naïve, treatment with pirtobrutinib was well tolerated and demonstrated NI to ibrutinib in IRC-assessed ORR, with numerically higher ORR, in both the ITT and R/R populations. Early trends in progression-free survival also favored pirtobrutinib, particularly in treatment-naïve patients.

Relevance *(J.W. Friedberg)*

These results demonstrate equivalent activity of pirtobrutinib compared with ibrutinib, and should inform potential use as upfront monotherapy for some patients with CLL, particularly with longer follow-up. Real-world studies evaluating sequencing are urgently needed to determine whether upfront pirtobrutinib is preferable to a standard second generation BTK inhibitor, versus reserving pirtobrutinib for the R/R situation.*

*Relevance section written by *JCO* Editor-in-Chief Jonathan W. Friedberg, MD.

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similar efficacy in high-risk relapsed patients, whereas zanubrutinib demonstrated improved overall response rate (ORR) and progression-free survival (PFS) in unselected relapsed CLL/SLL.[4,5] Second-generation cBTKi have not been compared head to head, nor compared with ibrutinib in the TN setting.[6,7]

Pirtobrutinib is a highly selective, noncovalent BTKi (ncBTKi) with low nanomolar potency, able to maintain continuous BTK inhibition throughout the dosing interval, and continues to inhibit BTK despite common BTKi resistance C481 mutations.[8] These properties suggest that pirtobrutinib may provide greater efficacy through more complete target inhibition while also delivering improved tolerability compared with cBTKis. In the phase I/II BRUIN study, pirtobrutinib demonstrated robust efficacy and a favorable safety profile in heavily pretreated patients with R/R CLL/SLL, including those previously exposed to cBTKi,[9] leading to accelerated approval by the US Food and Drug Administration (FDA) for adults with CLL/SLL who have received at least two prior lines of therapy, including a BTKi and a BCL2i.[10] The BRUIN CLL-321 phase III randomized-controlled trial validated pirtobrutinib's clinical benefit in patients previously treated with a cBTKi[11] and led to global approvals for adult patients with R/R CLL/SLL previously treated with a BTKi.

To investigate the potential for pirtobrutinib to deliver improved outcomes for patients with BTKi-naïve CLL/SLL, we conducted BRUIN CLL-314, to our knowledge, the first randomized, head-to-head trial comparing a ncBTKi, pirtobrutinib, with a cBTKi, ibrutinib, in patients with TN and R/R CLL. Here, we report the prespecified final analysis of ORR and descriptive interim analysis of PFS.

PATIENTS AND METHODS

Patients

Eligible patients who were TN or R/R without prior BTKi therapy, were age 18 years and older, and had a confirmed diagnosis of CLL/SLL requiring treatment per International Workshop on Chronic Lymphocytic Leukemia (iwCLL) 2018 criteria,[12] Eastern Cooperative Oncology Group performance status of 2 or less, and known deletion 17p status by fluorescence in situ hybridization (FISH) were enrolled. Recruitment of TN patients was formally halted once enrollment of this patient population approached the prespecified approximately 30% cap of total enrollment. Patients on concomitant anticoagulation (excluding vitamin K antagonists), receiving an antiplatelet agent, or a history of atrial fibrillation/atrial flutter that was not active at the time of enrollment, were permitted. Complete inclusion/exclusion criteria can be found in the protocol (online only).

Study Design and Treatment

BRUIN CLL-314 (LOXO-BTK-20030) is a global, multicenter, open-label, randomized phase III study of pirtobrutinib versus ibrutinib in patients with BTKi-naïve CLL/SLL (TN patients and patients with R/R disease). Patients were randomly assigned 1:1 by means of an interactive web-based response system to receive either pirtobrutinib or ibrutinib. Patients were stratified by del(17p) status (yes *v* no)

determined by FISH using local or central results and number of prior lines of therapy (0 *v* 1 *v* ≥2). All disease assessments performed by independent review committee (IRC) and investigator used iwCLL 2018 criteria. The study sponsor remained blinded to all aggregate, comparative analyses until the primary analysis of ORR.

Pirtobrutinib was administered at 200 mg orally once daily. Ibrutinib was administered at 420 mg orally once daily. Both treatments were administered continuously until disease progression, unacceptable toxicity, or other reasons for treatment discontinuation. Dose modifications and interruptions were allowed per protocol. No crossover was permitted.

Peripheral blood samples collected during screening or C1D1 were centrally tested by FISH, where allowed per country regulations (MPLN, Maryville, TN; Q2 Durham, NC; KingMed, Guangzhou, China) with either the Vysis CLL FISH Probe Kit (Abbott, Des Plaines, IL) or a laboratory developed test on the basis of FISH probes (Abbott at KingMed in China). *TP53* alterations were evaluated in peripheral blood mononuclear cells with the FoundationOne Heme assay (Foundation Medicine, Inc, Cambridge, MA). Immunoglobulin heavy chain variable region (IGHV) mutation status was evaluated in fresh peripheral blood, and screening bone marrow aspirates were evaluated centrally for karyotype via G-banding in stimulated culture (MPLN).

All site institutional review boards approved the trial protocol. The study was done according to the principles of the Declaration of Helsinki and Council for International Organizations of Medical Sciences International Ethical Guidelines. All patients provided written informed consent. The study is registered on ClinicalTrials.gov (identifier: NCT05254743).

Study End Points and Assessments

For regulatory filing with the FDA, the primary end point of ORR and the key secondary end point of PFS in the intention-to-treat (ITT) and R/R populations were assessed by a blinded IRC. For the primary end point, ORR was defined as the proportion of patients who achieve a best overall response of complete remission, complete remission with incomplete hematologic recovery, nodular partial remission (nPR), or PR at or before the initiation of subsequent anticancer therapy. Disease response was assessed in accordance with the 2018 criteria of the iwCLL,[12] every 4 months for 2 years, and every 6 months thereafter.

Other secondary end points included investigator-assessed ORR and PFS, IRC and investigator-assessed ORR that included PR with lymphocytosis (PR-L), overall survival (OS), and safety measures. Treatment-emergent adverse events (TEAEs) were defined as events that occurred or worsened in severity after baseline through 30 days (+7-day window) after the date of the last dose of study treatment or the first date of starting new anticancer therapy for CLL/SLL, whichever occurred earlier. Severity of TEAEs was graded per the National Cancer Institute Common Terminology Criteria in Adverse Events (CTCAE), Version 5.0. The assessment of exploratory patient-reported outcome version of the CTCAE (PRO-CTCAE) items is described in the Data Supplement (online only).

Statistical Analysis

With a sample size of 662 patients in the ITT population, there was 80% power for testing ORR noninferiority (NI) against a NI margin of 0.88 in the ORR ratio, assuming an ORR ratio of 1.03 (78% and 75.8% in pirtobrutinib and ibrutinib groups, respectively), at a two-sided significance level of .005. With a sample size of 437 in the R/R population, there was 90% power for testing ORR NI against a NI margin of 0.86 in the ORR ratio, assuming an ORR ratio of 1.05 (73.5% and 70% in pirtobrutinib and ibrutinib groups, respectively), at a two-sided significance level of .045.

To adjust for multiplicity and control the overall family-wise type I error rate at 0.05 (two-sided), a graphical testing strategy[13] was used for testing two primary hypotheses of IRC-ORR NI and two key secondary hypotheses of IRC-PFS superiority in ITT and R/R populations. The two-sided alpha levels initially allocated to test NI of IRC-ORR were .005 and .045 in the ITT and R/R populations, respectively. Since the NI criteria were met in both populations, the two-sided alpha level of .05 was passed to test the hypothesis of IRC-PFS superiority in the R/R population. If significant, superiority in IRC-PFS in the ITT population will be tested. At this prespecified ORR final analysis, however, interim PFS analyses were meant to be descriptive with a small technical alpha allocated, and remaining alpha will be spent at subsequent PFS analyses planned.

ORR NI hypotheses were tested with a stratified Wald test for the ORR ratio, and Wald CIs on the ORR ratio were provided. ORR was calculated along with Clopper-Pearson CI, and differences in ORR were evaluated using the stratified Cochran-Mantel-Haenszel test. In subgroup analyses, differences in ORR were calculated along with Wald CI. PFS and other time-to-event end points were estimated using the Kaplan-Meier method and differences tested with a stratified log-rank test. Hazard ratios (HRs) and Wald CI were estimated using the stratified Cox proportional hazards model. All stratified analyses included del(17p) as random assignment strata, and stratified analyses in the ITT and R/R populations also included number of prior lines of therapy (0 *v* 1 *v* ≥2, and 1 *v* ≥2, respectively) as random assignment strata. Further details regarding the graphical testing strategy, the test statistics used for evaluating NI and differences in ORR, and the exploratory analysis of PRO-CTCAE items are described in the Data Supplement.

All *P* values presented are two-sided, and all CIs are given at a two-sided 95% level (95% CI). *P* values from tests of

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secondary and exploratory end points, including analyses in the TN population, are not adjusted for multiplicity and are considered nominal. Statistical analyses were performed using SAS software (SAS 9.4, SAS Institute Inc, Cary, NC) and R software (R Core Team, version 4.4.2, Vienna, Austria). A data cutoff date of June 10, 2025, was used for all analyses.

RESULTS

Patient Population

From August 18, 2022, to June 17, 2024, 662 patients across 23 countries were randomly assigned to receive pirtobrutinib (n = 331) or ibrutinib (n = 331; ITT population). The ITT population included 225 TN patients (112 assigned to pirtobrutinib and 113 assigned to ibrutinib) and 437 patients with R/R disease (219 assigned to pirtobrutinib and 218 assigned to ibrutinib; Fig 1).

Baseline characteristics were generally balanced between groups (Table 1). Among all randomly assigned patients, the median age was 67 years (range, 34-90), 64.7% were male, and 52.1% were from Europe. For patients with R/R disease, the median number of prior lines of therapy was one in both the pirtobrutinib (range, 1-9) and ibrutinib (range, 1-8) groups. In the ITT population, the prevalence of high-risk molecular features was generally balanced between the pirtobrutinib and ibrutinib groups, including del(17p), and among patients with evaluable samples, unmutated IGHV status and complex karyotype (≥3 abnormalities).

One patient randomly assigned to pirtobrutinib and six patients randomly assigned to ibrutinib did not receive treatment. At the time of analysis, 78.5% (172/218) of patients in the R/R population remained on treatment with pirtobrutinib, compared with 70.2% (153/216) on ibrutinib. In the TN population, 86.6% (97/112) remained on pirtobrutinib and 68.1% (77/109) on ibrutinib. Among all randomly assigned patients, the most frequent reasons for treatment discontinuation were AEs (pirtobrutinib, 32 [9.7%]; ibrutinib, 35 [10.6%]) and progressive disease (pirtobrutinib, 15 [4.5%]; ibrutinib, 36 [10.9%]; Fig 1).

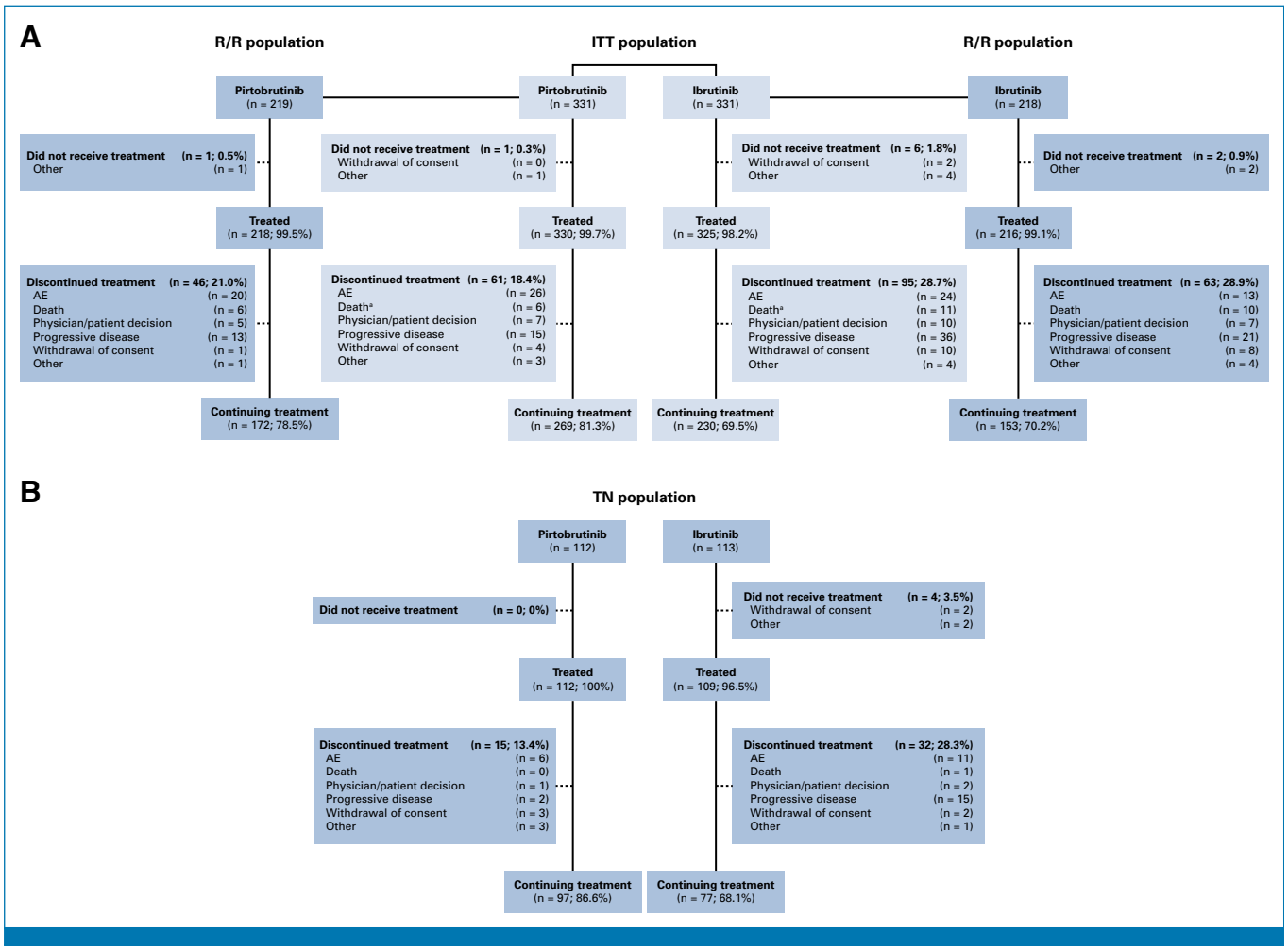

FIG 1. BRUIN CLL-314 study CONSORT diagram of the (A) ITT and R/R populations and (B) TN population. A total of 662 patients were randomly assigned. All 662 randomly assigned patients were included in the efficacy analyses. [a]All deaths were due to AE. AE, adverse event; ITT, intention to treat; R/R, relapsed/refractory; TN, treatment-naïve.

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TABLE 1. Demographic and Clinical Characteristics of the Patients at Baseline in the ITT and TN Populations

	ITT, No. (%)		R/R, No. (%)		TN, No. (%)	
Characteristic	Pirtobrutinib (n = 331)	Ibrutinib (n = 331)	Pirtobrutinib (n = 219)	Ibrutinib (n = 218)	Pirtobrutinib (n = 112)	Ibrutinib (n = 113)
Age, years, median (range)	67 (39-90)	67 (34-86)	67 (42-87)	67 (38-86)	67 (39-90)	68 (34-84)
≥65 years	194 (58.6)	199 (60.1)	126 (57.5)	133 (61.0)	68 (60.7)	66 (58.4)
Sex						
Male	213 (64.4)	215 (65.0)	145 (66.2)	144 (66.1)	68 (60.7)	71 (62.8)
Female	118 (35.6)	116 (35.0)	74 (33.8)	74 (33.9)	44 (39.3)	42 (37.2)
Region						
North America	26 (7.9)	21 (6.3)	12 (5.5)	13 (6.0)	14 (12.5)	8 (7.1)
Europe	174 (52.6)	171 (51.7)	107 (48.9)	108 (49.5)	67 (59.8)	63 (55.8)
Asia	42 (12.7)	51 (15.4)	27 (12.3)	32 (14.7)	15 (13.4)	19 (16.8)
Other[a]	89 (26.9)	88 (26.6)	73 (33.3)	65 (29.8)	16 (14.3)	23 (20.4)
Rai stage[b]						
0 to II	168 (54.9)	171 (58.2)	112 (53.8)	104 (55.0)	56 (57.1)	67 (63.8)
III to IV	135 (44.1)	119 (40.4)	94 (45.2)	82 (43.4)	41 (41.8)	37 (35.2)
Missing	3 (1.0)	4 (1.4)	2 (1.0)	3 (1.6)	1 (1.0)	1 (1.0)
Baseline ECOG PS						
0	176 (53.2)	170 (51.4)	118 (53.9)	107 (49.1)	58 (51.8)	63 (55.8)
1	143 (43.2)	151 (45.6)	94 (42.9)	104 (47.7)	49 (43.8)	47 (41.6)
2	12 (3.6)	10 (3.0)	7 (3.2)	7 (3.2)	5 (4.5)	3 (2.7)
Duration of disease, years, median (Q1, Q3)	5.62 (2.20, 8.91)	5.26 (1.88, 9.73)	7.29 (4.55, 10.75)	7.45 (3.84, 10.97)	1.97 (0.30, 4.50)	1.81 (0.46, 4.45)
Bulky disease[c]	107 (32.3)	116 (35.0)	69 (31.5)	83 (38.1)	38 (33.9)	33 (29.2)
Baseline β2 microglobulin						
≤3.5 mg/L	107 (32.3)	118 (35.6)	66 (30.1)	76 (34.9)	41 (36.6)	42 (37.2)
>3.5 mg/L	213 (64.4)	207 (62.5)	145 (66.2)	136 (62.4)	68 (60.7)	71 (62.8)
Missing	11 (3.3)	6 (1.8)	8 (3.7)	6 (2.8)	3 (2.7)	0 (0)
High-risk molecular features No./n available (%)						
IGHV unmutated[d]	199/293 (67.9)	183/277 (66.1)	141/194 (72.7)	129/184 (70.1)	58/99 (58.6)	54/93 (58.1)
17p deletion[e]	50/331 (15.1)	52/331 (15.7)	36/219 (16.4)	35/218 (16.1)	14/112 (12.5)	17/113 (15.0)
11q deletion[d]	57/255 (22.4)	48/226 (21.2)	47/169 (27.8)	42/155 (27.1)	10/86 (11.6)	6/71 (8.5)
TP53 mutation[d]	92/284 (32.4)	78/273 (28.6)	81/189 (42.9)	59/179 (33.0)	11/95 (11.6)	19/94 (20.2)
Complex karyotype[d,f]	104/259 (40.2)	78/227 (34.4)	76/168 (45.2)	60/150 (40.0)	28/91 (30.8)	18/77 (23.4)

Abbreviations: CLL, chronic lymphocytic leukemia; ECOG PS, Eastern Cooperative Oncology Group performance status; IGHV, immunoglobulin heavy chain variable region; ITT, intention to treat; SLL, small lymphocyctic lymphoma; R/R, relapsed/refractory; *TP53*, tumor protein p53; TN, treatment-naïve.

[a]Includes South America/Australia/New Zealand/Middle East.

[b]Percentages based on the number of participants with CLL (not SLL).

[c]Target lymph node ≥5 cm.

[d]Central laboratory test.

[e]Local or central laboratory test reported via electronic data capture.

[f]Defined as ≥3 chromosomal abnormalities in same clonal population.

Overall Response

Pirtobrutinib demonstrated NI to ibrutinib in IRC-ORR with statistical significance ($P < .0001$) in both the ITT (ORR ratio, 1.11 [95% CI, 1.03 to 1.19]) and R/R populations (ORR ratio, 1.12 [95% CI, 1.02 to 1.24]; Table 2). In the ITT population, IRC-ORR was higher with pirtobrutinib than with ibrutinib, 87.0% (95% CI, 82.9 to 90.4) *v* 78.5% (95% CI, 73.7 to 82.9; nominal $P = .0035$). Pirtobrutinib IRC-ORR was higher compared with ibrutinib in the R/R population (n = 437) at 84.0% (95% CI, 78.5 to 88.6) versus 74.8% (95% CI, 68.5 to 80.4; nominal $P = .0175$) and in TN patients (n = 225), 92.9% (95% CI, 86.4 to 96.9) versus 85.8% (95% CI, 78.0 to 91.7; nominal $P = .0886$). IRC-ORR including PR-L was also

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TABLE 2. IRC-Assessed ORR Across Populations

	ITT, No. (%)		R/R, No. (%)		TN, No. (%)	
BOR[a]	Pirtobrutinib (n = 331)	Ibrutinib (n = 331)	Pirtobrutinib (n = 219)	Ibrutinib (n = 218)	Pirtobrutinib (n = 112)	Ibrutinib (n = 113)
ORR (PR or better) ratio	1.1080		1.1233		1.0797	
95% CI	1.034 to 1.187		1.020 to 1.237		0.989 to 1.179	
P value for NI[b]	<.0001		<.0001		–	
ORR (PR or better)	288 (87.0)	260 (78.5)	184 (84.0)	163 (74.8)	104 (92.9)	97 (85.8)
95% CI[c]	82.90 to 90.44	73.73 to 82.85	78.48 to 88.61	68.46 to 80.39	86.41 to 96.87	78.03 to 91.68
P[d]	.0035		.0175		.0886	
CR or CRi	16 (4.8)	8 (2.4)	8 (3.7)	4 (1.8)	8 (7.1)	4 (3.5)
PR or nPR	272 (82.2)	252 (76.1)	176 (80.4)	159 (72.9)	96 (85.7)	93 (82.3)
PR-L	8 (2.4)	13 (3.9)	7 (3.2)	10 (4.6)	1 (0.9)	3 (2.7)
SD	18 (5.4)	36 (10.9)	15 (6.8)	31 (14.2)	3 (2.7)	5 (4.4)
PD	5 (1.5)	4 (1.2)	5 (2.3)	4 (1.8)	0 (0)	0 (0)
ORR including PR-L	296 (89.4)	273 (82.5)	191 (87.2)	173 (79.4)	105 (93.8)	100 (88.5)
95% CI[c]	85.60 to 92.52	77.95 to 86.42	82.05 to 91.33	73.37 to 84.53	87.55 to 97.45	81.13 to 93.73
P[d]	.0093		.0286		.1692	

NOTE. Pirtobrutinib demonstrated consistently higher ORR than ibrutinib across all populations evaluated, including ITT, R/R, and TN cohorts. ORR is defined as the proportion of patients who achieve a BOR of CR, CRi, nPR, or PR at or before the initiation of subsequent anti-cancer therapy. Abbreviations: BOR, best overall response; CR, complete remission; CRi, complete remission with incomplete hematologic recovery; IRC, Independent Review Committee; ITT, intention to treat; NI, noninferiority; nPR, nodular partial remission; ORR, overall response rate; PD, progressive disease; PR, partial remission; PR-L, partial remission with lymphocytosis; R/R, relapsed/refractory; SD, stable disease; TN, treatment-naïve.
[a]Does not include response categories non-PD, no evidence of disease, unknown, or not applicable.
[b]Two-sided *P* value based on the stratified Wald test for NI with NI margins of 0.88 and 0.86 in ITT and R/R populations. NI was also evaluated using CI: in the ITT population, the 99.5% CI was (1.004 to 1.223), with the lower limit excluding the NI margin, and in the R/R population, the 95.5% CI was (1.018 to 1.240), with the lower limit excluding the NI margin, thus meeting the prespecified criteria for NI.
[c]95% CI based on the exact binomial method (Clopper-Pearson).
[d]Two-sided nominal *P* value (Cochran-Mantel-Haenzel test) comparing response rates between groups.

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higher with pirtobrutinib compared with ibrutinib in the ITT (89.4% [95% CI, 85.6 to 92.5] *v* 82.5% [95% CI, 77.9 to 86.4]), R/R (87.2% [95% CI, 82.1 to 91.3] *v* 79.4% [95% CI, 73.4 to 84.5]) and TN (93.8% [95% CI, 87.5 to 97.5] *v* 88.5% [95% CI, 81.1 to 93.7) populations, respectively.

Investigator-assessed ORR was consistent with IRC findings (Data Supplement, Table S1). Investigator-assessed ORR was higher with pirtobrutinib than with ibrutinib across all populations: in the ITT population, 89.1% (95% CI, 85.3 to 92.3) versus 78.9% (95% CI, 74.1 to 83.1; nominal $P = .0003$); in the R/R population, 87.7% (95% CI, 82.6 to 91.7) versus 75.2% (95% CI, 68.9 to 80.8; nominal $P = .0009$); and in the TN population, 92.0% (95% CI, 85.3 to 96.3) versus 85.8% (95% CI, 78.0 to 91.7; nominal $P = .1480$). These results were consistent with ORR including PR-L.

IRC-ORR was generally higher with pirtobrutinib than with ibrutinib across prespecified subgroups, including patients with and without del(17p), as well as those with unmutated IGHV status and complex karyotype (Data Supplement, Fig S1). These results were generally consistent across the R/R and TN populations (data not shown).

Progression-Free Survival

Investigator-assessed PFS favored pirtobrutinib in the ITT (HR, 0.57 [95% CI, 0.39 to 0.83]), R/R (HR, 0.73 [95% CI, 0.47 to 1.13]), and TN (HR, 0.24 [95% CI, 0.10 to 0.59]) populations, with a median follow-up of 21.8, 18.2, and 22.5 months, respectively. At 18 months, PFS rates were 86.9% (95% CI, 82.4 to 90.3) and 82.3% (95% CI, 77.3 to 86.3) with pirtobrutinib and ibrutinib, respectively, in the ITT population, 81.7% (95% CI, 75.1 to 86.7) versus 79.2% (95% CI, 72.3 to 84.6) in the R/R population, and 95.3% (95% CI, 89.1 to 98.0) versus 87.6% (95% CI, 79.7 to 92.6) in the TN population (Figs 2A-2C). IRC PFS also trended in favor of pirtobrutinib in the ITT (HR, 0.76 [95% CI, 0.53 to 1.08]; Fig 2D), R/R (HR, 0.845 [95% CI, 0.57 to 1.26]), and TN (HR, 0.50 [95% CI, 0.23 to 1.08]) populations (Data Supplement, Fig S2), at the time of the interim analysis for PFS.

Overall Survival

At data cutoff, deaths were observed in 7.6% (25/331) and 7.3% (24/331) of patients in the ITT population, 10.0% (22/219) and 9.6% (21/218) in the R/R population, and 2.7%

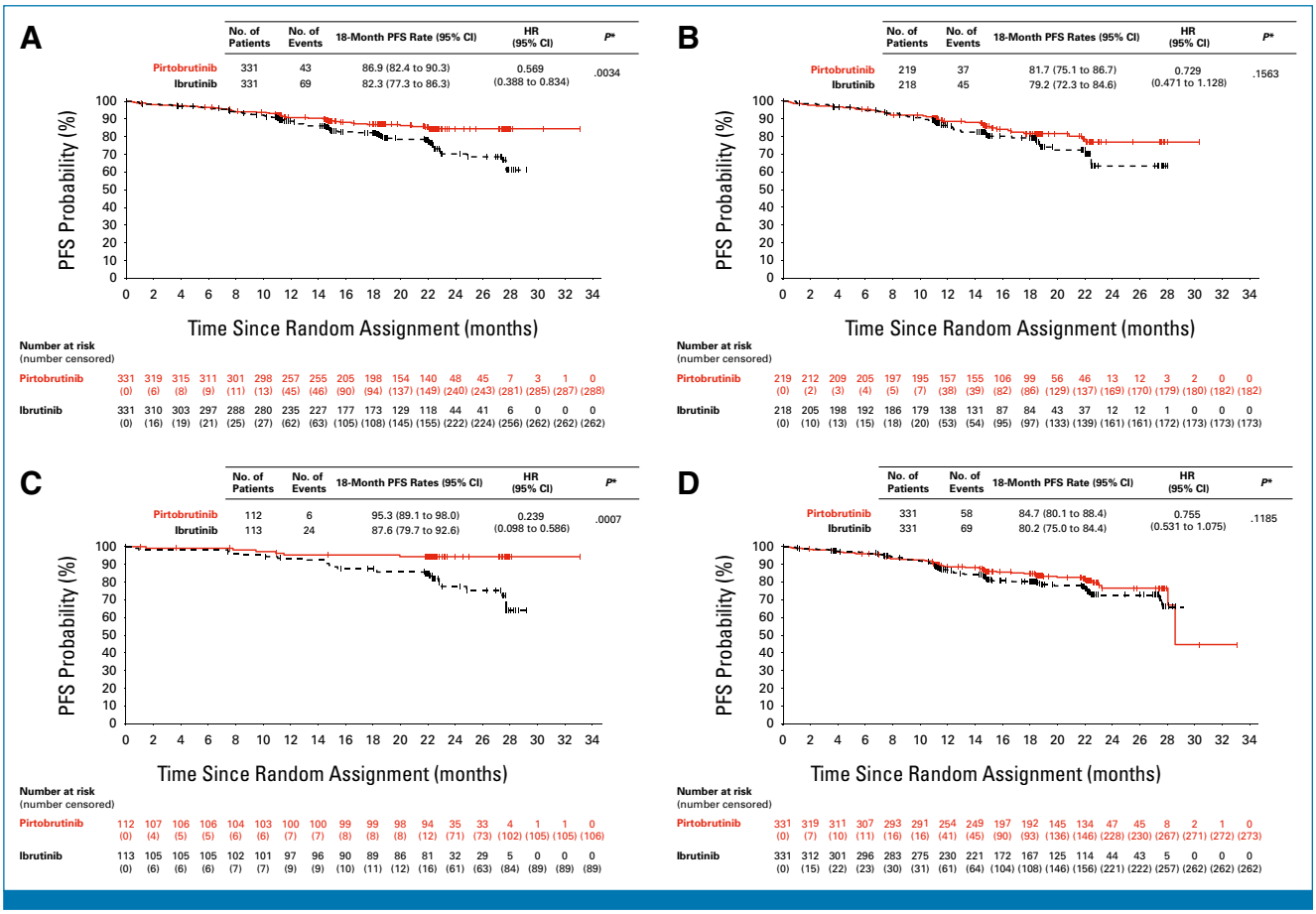

FIG 2. Kaplan-Meier curve of PFS assessed by investigator in the (A) ITT, (B), R/R, and (C) TN populations and by IRC in the (D) ITT population. *Two-sided *P* values are based on the stratified log-rank test and are reported as nominal values for investigator-assessed PFS. HR, hazard ratio; IRC, independent review committee; ITT, intention to treat; PFS, progression-free survival; R/R, relapsed/refractory; TN, treatment-naïve.

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(3/112) and 2.7% (3/113) in the TN population, in the pirtobrutinib and ibrutinib groups, respectively. OS was immature (ITT HR, 0.96 [95% CI, 0.55 to 1.69]; R/R HR, 0.96 [95% CI, 0.53 to 1.76]; TN HR, 0.99 [95% CI, 0.20 to 4.89]).

Safety

Safety was evaluated in all patients who received at least one dose of pirtobrutinib or ibrutinib (n = 655; Table 3). The median treatment duration with pirtobrutinib was 20.5 months (IQR, 15.18-26.51) and with ibrutinib was 19.3 months (IQR, 13.37-25.99). The incidence of TEAEs of any grade was comparable between groups (97.0% for pirtobrutinib *v* 97.8% for ibrutinib), as was the rate of grade ≥3 TEAEs (54.8% *v* 53.5%). The TEAE of lymphocytosis occurred infrequently, in 3.6% of patients who received pirtobrutinib and 5.8% of patients who received ibrutinib. Although the median time to onset was approximately 4 weeks in both treatment groups, the median time to resolution appeared shorter with pirtobrutinib (Data Supplement, Table S2) Overall, safety was consistent across the R/R and TN populations (Data Supplement, Table S3). Grade 5 TEAEs were similar between groups (Data Supplement, Table S4).

AEs of interest are shown in Table 3. Atrial fibrillation/atrial flutter were less frequent with pirtobrutinib (2.4%; grade ≥3, 0.9%) compared with ibrutinib (13.5%; grade ≥3, 4.0%). Hypertension was also less frequent with pirtobrutinib (10.6% *v* 15.1; grade ≥3, 3.3% *v* 4.9%).

Patient-Reported Outcomes

The baseline-adjusted percentage of participants with worst patient-reported experience of symptomatic AEs was numerically lower during treatment with pirtobrutinib compared with ibrutinib in an exploratory analysis of the safety population (in terms of occurrence, frequency, severity, and/or interference with daily activities; Data Supplement, Table S5).

DISCUSSION

In this global randomized phase III study comparing a ncBTKi with a covalent BTKi in CLL/SLL, the primary end point was met, and pirtobrutinib demonstrated NI in IRC-ORR compared with ibrutinib. Additionally, the ORR was numerically higher with pirtobrutinib, and although immature, early trends in PFS also favored pirtobrutinib among

TABLE 3. Summary of Adverse Events in the Overall Safety Population (n = 655)

	Pirtobrutinib (n = 330), No. (%)		Ibrutinib (n = 325), No. (%)	
AE	Any Grade	Grade ≥3	Any Grade	Grade ≥3
Any TEAE	320 (97.0)	181 (54.8)	318 (97.8)	174 (53.5)
TEAEs present in ≥10% of patients in either treatment group[a]				
Upper respiratory tract infection	59 (17.9)	2 (0.6)	63 (19.4)	0
Pneumonia	45 (13.6)	21 (6.4)	49 (15.1)	28 (8.6)
Diarrhea	44 (13.3)	1 (0.3)	62 (19.1)	4 (1.2)
COVID-19	40 (12.1)	4 (1.2)	33 (10.2)	5 (1.5)
Contusion	33 (10.0)	0	30 (9.2)	0
Arthralgia	26 (7.9)	0	41 (12.6)	0
Urinary tract infection	26 (7.9)	3 (0.9)	40 (12.3)	3 (0.9)
TEAEs leading to treatment discontinuation[b]	31 (9.4)	25 (7.6)	35 (10.8)	25 (7.7)
AE of interest				
Anemia[c]	51 (15.5)	20 (6.1)	51 (15.7)	12 (3.7)
Atrial fibrillation and atrial flutter[d]	8 (2.4)	3 (0.9)	44 (13.5)	13 (4.0)
Bleeding	115 (34.8)	11 (3.3)	118 (36.3)	9 (2.8)
Bruising[e]	45 (13.6)	0	39 (12.0)	0
Petechiae and purpura	17 (5.2)	0	25 (7.7)	0
Hemorrhage[f]	78 (23.6)	11 (3.3)	81 (24.9)	9 (2.8)
Hypertension	35 (10.6)	11 (3.3)	49 (15.1)	16 (4.9)
Infections[g]	226 (68.5)	56 (17.0)	241 (74.2)	54 (16.6)
Infection without COVID-19	214 (64.8)	53 (16.1)	234 (72.0)	49 (15.1)
Neutropenia[h]	103 (31.2)	83 (25.3)	76 (23.4)	57 (17.5)
Thrombocytopenia[i]	39 (11.8)	12 (3.6)	57 (17.5)	13 (4.0)

Abbreviations: AE, adverse event; TEAE, treatment-emergent adverse event.
[a]Excluding hypertension, neutropenia, anemia, and thrombocytopenia as they are included under AE of interest as a preferred term or composite.
[b]In addition, one patient discontinued treatment in the pirtobrutinib arm because of AE that did not meet the definition of TEAE.
[c]Includes anemia, blood loss anemia, iron deficiency anemia, and microcytic anemia.
[d]Among patients ≥75 years, any-grade events occurred in 4.5% with pirtobrutinib versus 21.4% with ibrutinib; grade ≥3 events in 1.5% versus 7.1%, respectively.
[e]Includes contusion, ecchymosis, increased tendency to bruise, genital contusion, injection site bruising, oral contusion, and vessel puncture site bruise.
[f]Includes hemorrhage, hematoma, hematuria, hematospermia, epistaxis, coital bleeding, gingival bleeding, aortic rupture, blood loss anemia, oral blood blister, nail bed bleeding.
[g]Includes all infection events reported including COVID-19.
[h]Includes neutropenia, neutrophil count decrease, and febrile neutropenia.
[i]Includes thrombocytopenia and platelet count decrease.

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all patients and in the R/R and TN subgroups. Although informative censoring can introduce bias, the difference in early withdrawals without receiving treatment (1 *v* 6) represents <1% of the total enrollment (662 patients). Given this small imbalance and the robust demonstration of NI, there should be little impact of informative censoring on the primary result and other efficacy results.

The TN subgroup (n = 225) had the largest observed PFS treatment effect and the longest follow-up. This finding is both novel and clinically meaningful, as BTKis have become a cornerstone of treatment in patients with previously untreated CLL/SLL, but until now have not been investigated head to head in the TN setting as part of a randomized phase III trial. These data build on overall data demonstrating the efficacy of pirtobrutinib in CLL/SLL across the full continuum of care from patients who are TN, those who have received non-BTKi containing prior therapy, and those who relapsed or progressed after cBTKi exposure.

Pirtobrutinib demonstrated a favorable safety profile compared with ibrutinib. This finding was further supported by PROs, specifically PRO-CTCAE–based symptomatic AEs (myalgia, bruising, headache, diarrhea, and cough), which were identified a priori. Off-target kinase inhibition with cBTKi has been linked to AEs such as atrial fibrillation and hypertension, which frequently necessitate dose reductions or treatment discontinuation.[14,15] In our study, rates of atrial fibrillation and hypertension were lower with pirtobrutinib compared with ibrutinib, consistent with the higher

selectivity of pirtobrutinib for BTK. These improvements in tolerability are particularly important in CLL given the cardiac toxicities associated with cBTKi and together with the observed efficacy, suggesting that pirtobrutinib may address two major limitations of ibrutinib: the emergence of resistance and the burden of treatment-related toxicity. By achieving better disease control with favorable tolerability, pirtobrutinib may extend the duration of benefit from BTK inhibition and improve overall patient outcomes compared with a cBTKi.

The efficacy of pirtobrutinib in TN patients also raises important considerations for sequencing in CLL. The benefit of pirtobrutinib after having been treated with a cBTKi has already been demonstrated, although whether cBTKi inhibition is effective after progression on pirtobrutinib remains an open question. The results reported here suggest that the encouraging first-line efficacy and tolerability of pirtobrutinib may position it as a preferred BTKi option for safe long-term therapy among some TN patients. Venetoclax-based fixed-duration regimens offer an important alternative following relapse after continuous BTK inhibition. Prior data have demonstrated that venetoclax following pirtobrutinib remains an efficacious sequencing strategy.[16] For patients who may only need one or possibly two lines of therapy, such as those who are older at treatment initiation, sequencing considerations may have less clinical relevance, and physicians and patients may favor the treatment that offers the best tolerability and ease of administration.

This study has several limitations. The median follow-up remains limited for PFS, and further prespecified analyses are planned. This was an open-label study, and therefore, there is a risk of investigator bias. Importantly, a blinded IRC was implemented to assess response data, and the study sponsor remained blinded to all aggregate, comparative analyses until the primary analysis of ORR was conducted to reduce potential bias. The data generated in this study raise important considerations for sequencing of therapy in patients with CLL/SLL given the chronicity of this disease. Pirtobrutinib is the first ncBTKi approved and the first BTK inhibitor of any kind evaluated in a randomized phase III study in CLL/SLL patients with R/R disease all of whom were previously treated with a cBTKi.[11]

In summary, this novel phase III clinical trial demonstrates the potential role of the ncBTKi pirtobrutinib in treatment settings where covalent BTK inhibitors are currently considered standard of care.

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DISCLAIMER

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DATA SHARING STATEMENT

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Eli Lilly provides access to all individual participant data collected during the trial, after anonymization, except for pharmacokinetic or genetic data. Data are available to request 6 months after the indication studied has been approved in the United States and EU and after primary publication acceptance, whichever is later. No expiration date of data requests is currently set once data are made available. Access is provided after a proposal has been approved by an independent review committee identified for this purpose and after receipt of a signed data sharing agreement. Data and documents, including the study protocol, statistical analysis plan, clinical study report, and blank or annotated case report forms, will be provided in a secure data-sharing environment. For details on submitting a request, see the instructions provided at www.vivli.org.

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Pirtobrutinib Versus Ibrutinib in Treatment-Naïve and Relapsed/Refractory Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma

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No other potential conflicts of interest were reported.

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